

NATIONAL ECONOMICS UNIVERSITY

Faculty of Data Science and Artificial Intelligence

STATISTICS

Machine Learning-Based Electricity Consumption Forecasting in Tetouan City

A chronological evaluation of conventional regression models

GROUP MEMBERS

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Can machine learning beat a strong daily pattern?

PROBLEM MOTIVATION & RESEARCH QUESTIONS

Electricity planning needs the **shape of tomorrow's hourly demand**.

The practical reference

If demand repeats every day, copying yesterday is already competitive.

1. How much do conventional models improve on seasonal baselines?
2. Which feature groups add predictive value?
3. How do tuning and the target formulation affect the result?

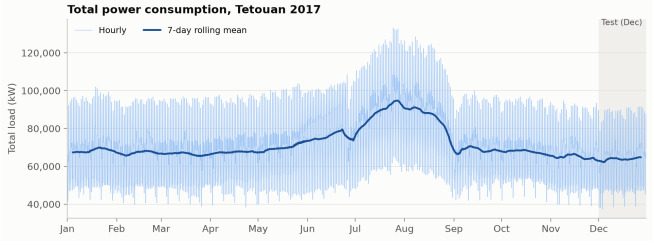
Evaluate future periods and interpret gains against the named baseline.

The target is mean hourly power across three zones

$$y_t = \frac{1}{6}\sum_{j=0}^{5}\left(Z_{1,t,j} + Z_{2,t,j} + Z_{3,t,j}\right) \quad [\mathrm{kW}]$$

Z1–Z3 form the target only, never the inputs.

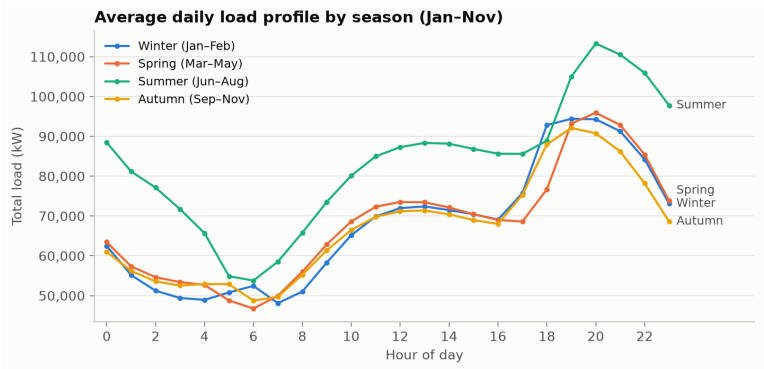
52,416 × **9**: 10-minute records
timestamp, 5 weather, 3 zone loads
8,736 hourly means, 2017



Mean power (kW); hourly energy is $y_t \times 1\,\mathrm{h}$ (kWh).

Whole-year view includes December, but no 31 December data; the centred smooth is descriptive.

Daily shape and season both matter



JANUARY–NOVEMBER

06–07h
daily minimum

19–20h
evening peak

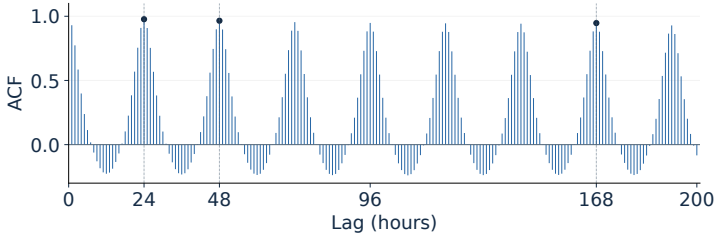
Summer
higher load profile

→ **features: `hour`, `hour_sin`, `hour_cos`; temperature for the seasonal level.**

Profiles use January–November development data.

Today looks like yesterday

JANUARY–NOVEMBER DATA ONLY



ACF: **24 h = 0.98**, **48 h = 0.96**, **168 h = 0.95** (weekly peak).

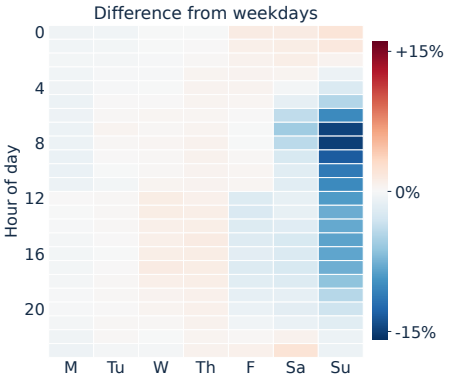
Lag 12 h is negative: day and night are in opposite phases.

Copying yesterday (Naive-24h) is therefore a strong baseline.

→ **features: `lag_24`, `lag_48` (daily history).**

Sundays and Mondays break the daily pattern

JANUARY–NOVEMBER / DIFFERENCE FROM MON–FRI AT THE SAME HOUR



Sundays: **5.9% lower** on average; about **15% lower at 7–8 h**.

Saturday mornings: **4–6% lower**.

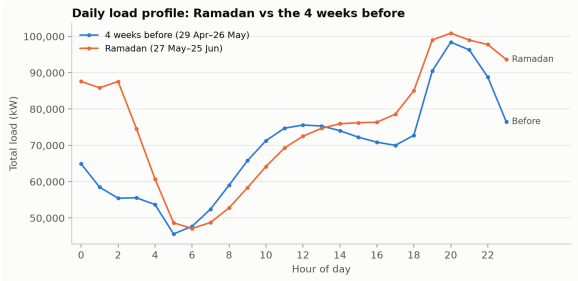
The effect varies by hour; a nonlinear model can learn this interaction.

Copying yesterday mismatches **Monday/Sunday** and **Sunday/Saturday**.

→ **features:** `dayofweek`, `is_weekend`, **cyclic weekday.**

Holidays and Ramadan move the load

JANUARY–NOVEMBER DATA ONLY



Religious holidays (two Eids, Islamic New Year):
3–13% lower than the surrounding days, about -9% on average.

Civil holidays: weaker and uneven, about -3%.
$\Rightarrow$ two separate flags, not one.

Ramadan: night load jumps (**+32,000 kW at 02:00**);
temperature differs by only 1.9 °C.

$\rightarrow$ **features:** `is_religious_holiday`, `is_civil_holiday`, `is_ramadan`.

Dates come from the official calendar, known in advance.

h24 and h1 use different information at issuance

h24: midnight batch

00:00 on day D: forecast its 24 hourly intervals.

History: $y_{t-24},\ y_{t-48}$.

Lead to interval start: 0–23 h.

Lead to interval completion: 1–24 h.

h1: hourly update

Update each hour for the next complete hourly interval.

History: $y_{t-1}, y_{t-2}, y_{t-3}, y_{t-24}$.

Uses the latest completed-hour load.

Prompt measurements assumed.

The 24-hour lag is a minimum history offset; lead times vary across the daily batch.

Both settings assume prompt load measurements and use observed target-hour weather.

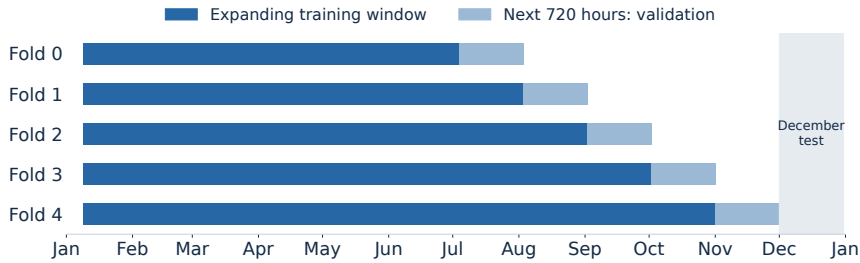
Seventeen inputs combine timing, weather and history

MAIN h24 MODEL: FEATURE SET D = 5 ORIGINAL + 12 ENGINEERED

Group	Inputs	Built from	Count
In the original dataset			
Weather	Temperature, humidity, wind speed, 2 diffuse-radiation flows	raw columns	5
Engineered by us			
Calendar	Hour, weekday, weekend flag; sin/cos of hour and weekday	timestamp	7
Special days	Religious holiday, civil holiday, Ramadan	2017 calendar	3
Daily history	y_{t-24}, y_{t-48}	past total load	2

Model selection stays within January–November

JAN–NOV DEVELOPMENT: 7,825 HOURS **DECEMBER TEST: 720 HOURS**



After history warm-up, five expanding windows each validate the next 30 days.

Selection: lowest mean CV RMSE → refit on all development data → December.

Tuning and reporting reuse folds (no nested CV); zero gap assumes prompt data.

Learn the load level—or a correction to yesterday

TWO SEASONAL BASELINES

$$\widehat{y}_t = y_{t-24} \qquad \widehat{y}_t = y_{t-168}$$

FIVE REGRESSION FAMILIES

Linear regression / tuned Ridge
Decision tree (DT)
Random forest (RF)
RBF support vector regression (SVR)
XGBoost (XGB)

TWO TARGET FORMULATIONS

Absolute: learn y_t.

Residual (Δ):

$$r_t = y_t - y_{t-24}$$

$$\widehat{y}_t = y_{t-24} + g(\mathbf{x}_t)$$

Learn the change from yesterday so recent load carries the current level.

Both target formulations compete on development CV.

RF averages are range-bounded; boosted trees do not share this strict bound.

Why predict a correction? July tests extrapolation

FOLD 0: 4 JULY–2 AUGUST / TUNED MODELS

A new load range

Training maximum: **110,711 kW**
Validation maximum: **133,130 kW**
About **12%** of hours exceed the training maximum.
For $r_t = y_t - y_{t-24}$, **0%** fall outside the training residual range.

July-fold RMSE (kW)

Model	Absolute	Residual
RF	5,554	**3,215**
XGBoost	5,344	**2,979**

RF averages cannot extrapolate beyond training targets. Learning a correction reduces the impact of the level shift.

Residual learning is not necessary for the selected SVR.

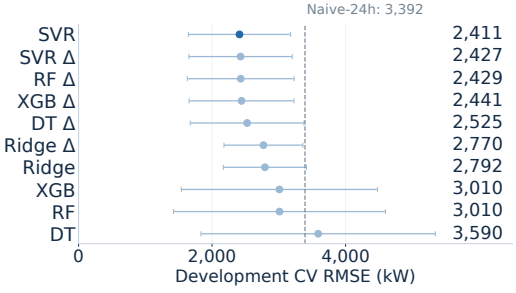
SVR CV: absolute **2,411** vs residual **2,427 kW**; nearly tied.

Target variants were tuned separately; gains also reflect their chosen parameters.

Absolute SVR leads CV, with close alternatives

Randomized search: **170 configurations** × **5 temporal folds per target type**;

SVR CV RMSE fell **39.3%** versus default SVR.



2,410.6

kW mean CV RMSE

± 766.3 kW fold SD.

Default models: XGB wins November; LR (**2,792 kW**) wins five-fold CV.

SVR and DT trail Naive-24h (**3,392 kW**) on CV.

More recent history: h24 vs h1 (CV). SVR 2,410.6 → 1,866.4 kW (-22.6%).

Same configurations/dates; histories and initial rows differ. CV folds are reused.

Daily lags drive accuracy; the expanded set hurts

ABSOLUTE-TARGET RF / FIVE-FOLD CV / FIXED D-TUNED PARAMETERS

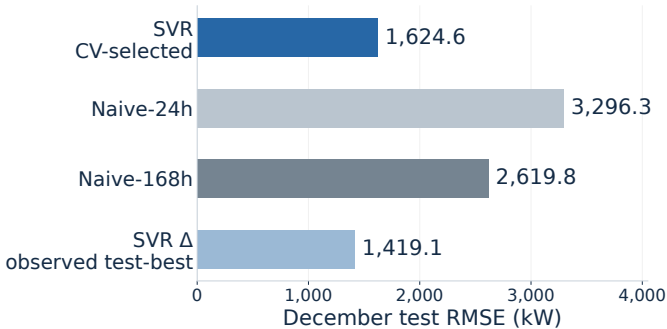
Set	Feature group	Inputs	RMSE (kW)	Change
A	Weather only	5	15,010	—
B	+ Calendar	12	10,041	-33.1%
C	+ Special days	15	9,168	-8.7%
D	**+ Daily lags (24 h, 48 h)**	**17**	**3,010**	-67.2%
E	+ Weekly lag and rolling summaries	21	3,700	$+22.9\%$

D is selected: daily lags give the largest gain; weekly/rolling features worsen RF and XGBoost (XGB **3,010** $\rightarrow$ **4,921 kW**), so they are **excluded from the final model**.

Set F = E without weather: RMSE $+1.6\%$ (RF), $+1.9\%$ (XGB), at most $+5.3\%$ (RF Δ).

F is not a test of the selected SVR.

The selected SVR halves daily-baseline December RMSE



50.7%

RMSE reduction vs Naive-24h

38.0% vs Naive-168h

MAE: **1,094.6 kW**

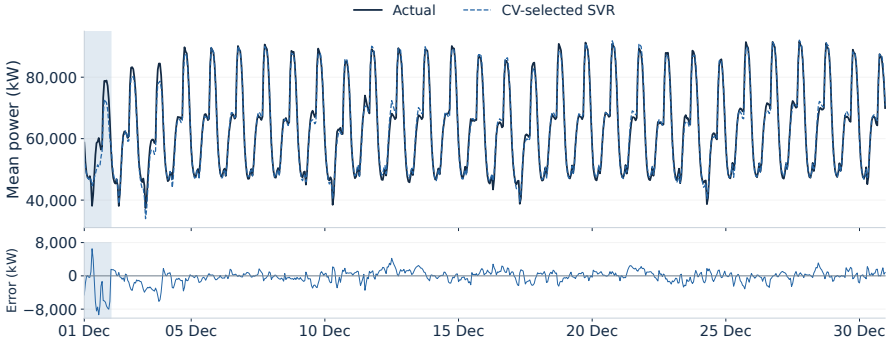
MAPE: **1.78%**

R^2: **0.9869**

Residual SVR: observed test-best; not CV-selected. Absolute SVR remains final.

December 1–30 only; results condition on observed target-hour weather.

Good daily tracking still leaves concentrated errors



Bias: -202.1 kW

Error = forecast − actual.

Worst day: 1 December (Mawlid)

MAE 4,579.2 kW; highlighted in blue.

Possible explanation: the holiday flag may transfer larger Eid demand reductions to Mawlid.

Compared with published Tetouan studies

VALUES AS REPORTED BY EACH STUDY, NOT RE-RUN

Study	Target	Forecast horizon	Split	MAPE	R^2
RoSe et al. (2023)	Total	none (no past load)	70/30, order not stated	–	0.976
Islam et al. (2025)	not stated	10 min ahead	75/15/10 chronological	6.3%	0.853
Alsalem (2025)	Zone 3 only	10 min ahead	70/30, order kept	–	0.989
Biswas et al. (2026)	mean of 3 zones	not stated	chronological	1.04%	–
This study	**Total**	**24 h (day-ahead)**	**chronological**	**1.78%**	**0.987**

Longer horizon: others forecast 10 minutes ahead, use no past load, or state no horizon.

Strict evaluation: model chosen on chronological CV; December used only once.

Named reference: −50.7% RMSE versus copying yesterday (Naive-24h).

Settings differ: a positioning, not a ranking. Salam & El Hibaoui (2018) and da Silva & Meneses (2023) report no MAPE or R^2.

Findings & limitations

FINDINGS

1. Choose the final model on chronological CV; report December separately.
2. Daily lags give the largest RF gain; weekly/rolling features hurt.
3. The CV-selected SVR halves December Naive-24h error (-50.7%).

LIMITATIONS

- Weather is observed at the target hour; a real midnight forecast needs weather forecasts.
- Prompt recent-load measurements assumed.
- One city, one year, one 30-day test.

Next: issuance-time weather and independent seasonal tests.

Conclusion & Q&A

CV-SELECTED ABSOLUTE-TARGET SVR / DECEMBER 2017

1,624.6 kW
RMSE on 720 hourly observations

50.7%
lower RMSE than Naive-24h

A promising chronological benchmark, conditional on observed weather.

Thank you. Questions?